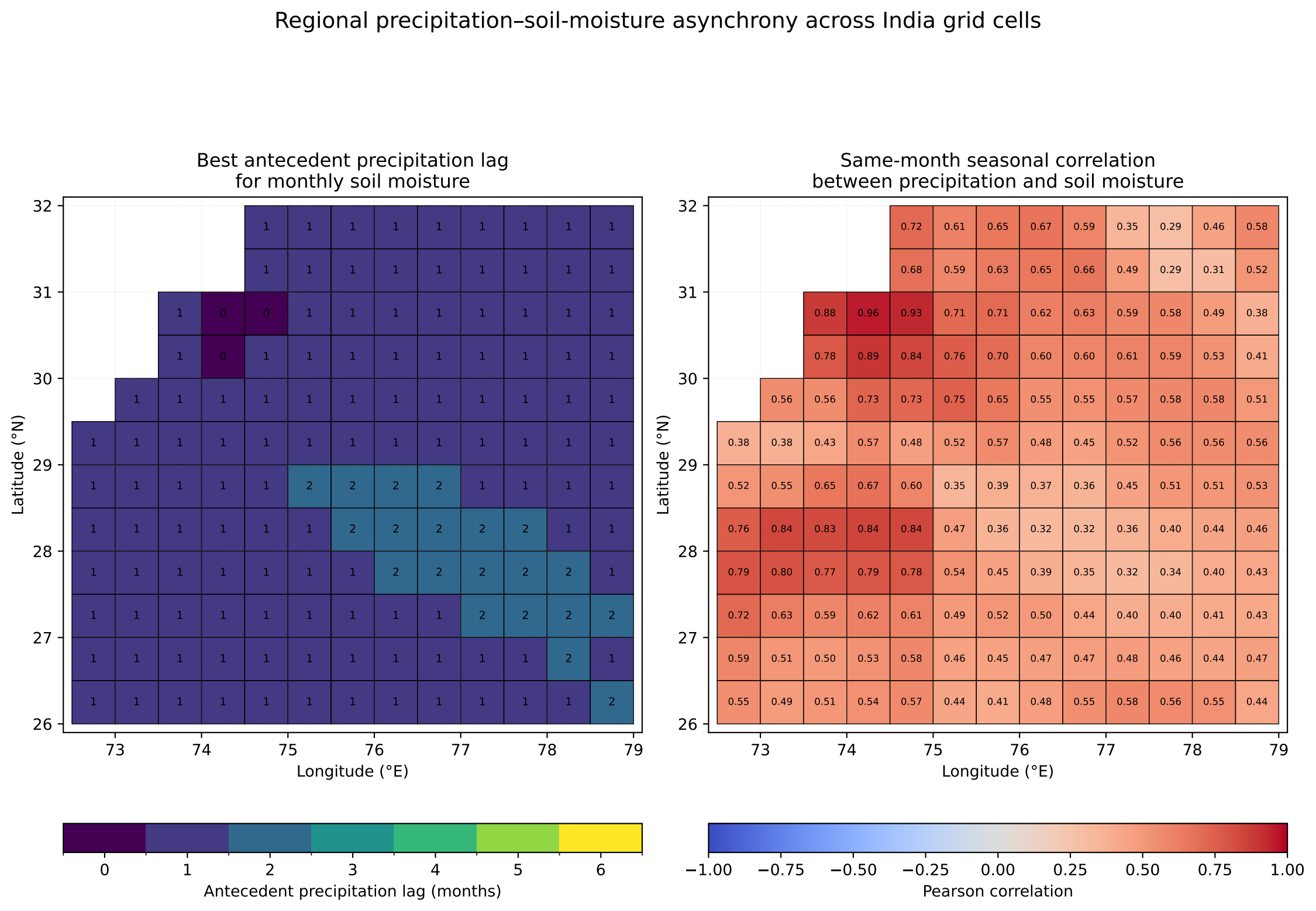
Regional precipitation–soil-moisture asynchrony across India grid cells
Best antecedent precipitation lag
for monthly soil moisture
Same-month seasonal correlation
between precipitation and soil moisture
Latitude (°N)
Longitude (°E)
32
31
30
29
28
27
26
73
74
75
76
77
78
79
1 1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1 1
1 0 0 1 1 1 1 1 1 1 1
1 0 1 1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1 1 1 2 2 2 2 1 1 1 1
1 1 1 1 1 1 2 2 2 2 2 1 1
1 1 1 1 1 1 1 2 2 2 2 2 1
1 1 1 1 1 1 1 1 1 2 2 2 2
1 1 1 1 1 1 1 1 1 1 1 2 1
1 1 1 1 1 1 1 1 1 1 1 1 2
0.72 0.61 0.65 0.67 0.59 0.35 0.29 0.46 0.58
0.68 0.59 0.63 0.65 0.66 0.49 0.29 0.31 0.52
0.88 0.96 0.93 0.71 0.71 0.62 0.63 0.59 0.58 0.49 0.38
0.78 0.89 0.84 0.76 0.70 0.60 0.60 0.61 0.59 0.53 0.41
0.56 0.56 0.73 0.73 0.75 0.65 0.55 0.55 0.57 0.58 0.58 0.51
0.38 0.38 0.43 0.57 0.48 0.52 0.57 0.48 0.45 0.52 0.56 0.56 0.56
0.52 0.55 0.65 0.67 0.60 0.35 0.39 0.37 0.36 0.45 0.51 0.51 0.53
0.76 0.84 0.83 0.84 0.84 0.47 0.36 0.32 0.32 0.36 0.40 0.44 0.46
0.79 0.80 0.77 0.79 0.78 0.54 0.45 0.39 0.35 0.32 0.34 0.40 0.43
0.72 0.63 0.59 0.62 0.61 0.49 0.52 0.50 0.44 0.40 0.40 0.41 0.43
0.59 0.51 0.50 0.53 0.58 0.46 0.45 0.47 0.47 0.48 0.46 0.44 0.47
0.55 0.49 0.51 0.54 0.57 0.44 0.41 0.48 0.55 0.58 0.56 0.55 0.44
0
1
2
3
4
5
6
Antecedent precipitation lag (months)
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Pearson correlation